

BIMM-143: INTRODUCTION TO BIOINFORMATICS

The find-a-gene project assignment

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Questions:

[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as **it's function is known**.

If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: ATP-binding cassette sub-family G member 1 isoform 7

Accession Number: NP_997512

Species: Homo sapiens

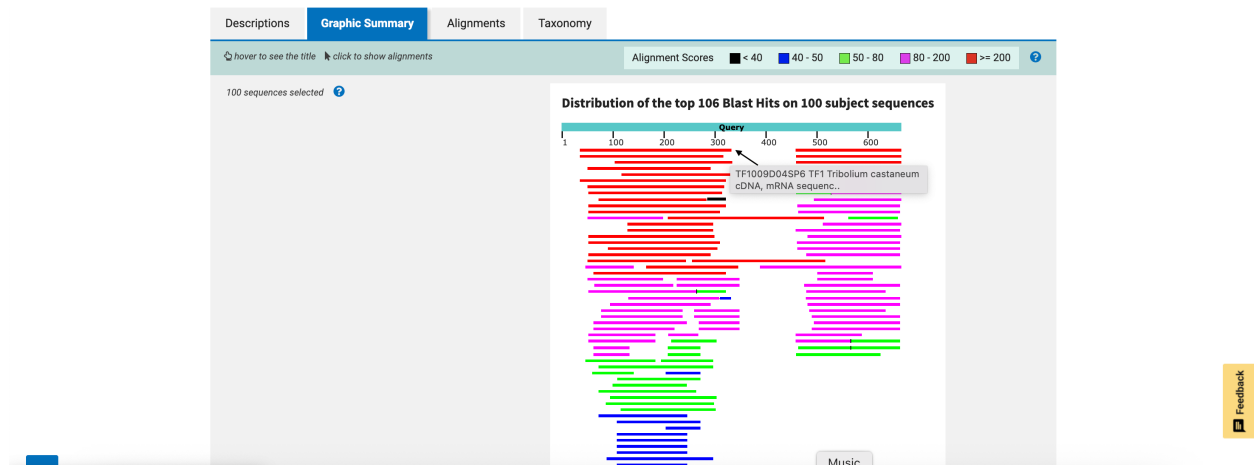
Function: ABCG1 is a transport protein involved in the efflux of phospholipids (including cholesterol) from cells, thus contributing to lipid/cholesterol homeostasis.

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

BLAST method: TBLASTN 2.17.0+ search against red flour beetle ESTs

Database searched: Expressed Sequence Tags (est)

Limits applied (Species): Red flour beetle (Taxid: 7070)



On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen match: Accession [ES547654.1](#), a 1025 base pair mRNA sequence from *Tribolium castaneum* (red flour beetle). Alignment details in the screenshot and text below.

Descriptions	Graphic Summary	Alignments	Taxonomy
Alignment view: Pairwise Restore defaults Download			
100 sequences selected			
Download GenBank Graphics Next Previous Descriptions			
TF1009D04SP6 TF1 Tribolium castaneum cDNA, mRNA sequence Sequence ID: ES547654.1 Length: 1025 Number of Matches: 1			
Range 1: 113 to 997 GenBank Graphics Next Match Previous Match			
Score	Expect	Method	Identities
326 bits(835)	3e-106	Compositional matrix adjust.	169/302(56%)
Query	37	NMEATETDLLGHLK---KVDNNLTEAQRFSLLPRAAANIIEFRDLSYSVPPEGWRRKGGY	94
Sbjct	113	NKMATDTVLLNRRSSDYKVOITPTQOKLTHLPQRPKVDLAFSNLKYTVKQG-----KAE	277
Query	95	KTLLKGISGKFNSELVAIMGPSGAGKSTLNNILAGYRETQMKGAVLINGLPRDLRCFRK	154
Sbjct	278	KT+LK+SG+ SGEL AIMGPSGAGKSTL+NIL GYR G++G +L+N RDL FRK	457
Query	155	VSCYIMQDDMLPHLTVOEAMMVSAHLKLOEKDEGRE-MVKEILTALGLLSCANTRTGS	213
Sbjct	458	LSAYIMQDNLHLNLTVDAMNVAALKIIGESKSEREDIIITEILDTLGLDHRKMTMSG	637
Query	214	LSGGQKRLAIALELVNPPVPMFDEPTSGLDSCFVVSMLGGLAAGGASTICTIHOP	273
Sbjct	638	LSGGQ+KRL+IALELV+NPP+MFDEPTSGLD+SCFQ +SL+K LA+GGR+IICTIHOP	817
Query	274	SAKLFEFDQLYVLSQGQCVYRGKVNQLVPYLRDLGLNCPITYN---PADFVMEVASGEY	330
Sbjct	818	SARLTFEPDHLTYLADGQCVYQSTKQLVPFLATLG---TPMXHRTXPXSYYIXVAGEY	991
Query	331	GD 332	
Sbjct	992	GD 997	

Alignment details (restated from screenshot above):

>TF1009D04SP6 TF1

Tribolium castaneum cDNA, mRNA sequence

Sequence ID: ES547654.1

Length: 1025

Range 1: 113 to 997

Score = 326 bits (835), Expect = 3e-106, Method: Compositional matrix adjust.

Identities = 169/302 (56%), Positives = 212/302 (70%), Gaps = 13/302(4%), Frame = +2

```
Query   37   NMEATETDLLNGHLK--KVDNNLTEAQRFSSLPRRAAVNIEFRDLSYSVPEGPWRRKKGY   94
          N  AT+T LL          KV      T+ +  + LP+R  V++ F +L Y+V +G      K
Sbjct   113  NKMATDTVLLKNRSSDYKVQITPTQQKTLTHLPQRPKVDLAFSNLKYTVKQG-----KAE   277

Query   95   KTLLKGISGKFNSGELVAIMGPSGAGKSTLMNILAGYRETGMKGAVLINGLPRDLRCFRK   154
          KT+LK +SG+  SGEL AIMGPSGAGKSTL+NIL GYR  G++G +L+N   RDL  FRK
Sbjct   278  KTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYRTDGVQGQILMNDSELDLSQFRK   457

Query   155  VSCYIMQDDMLLPHLTVQEAMMVSAHLKLQEKDEGRRE-MVKEILTALGLLSCANTRTGS   213
          +S YIMQD+ L  +LTV EAM V+A LK+ EK +  RE ++ EIL  LGLL      T T
Sbjct   458  LSAYIMQDNQLHLNLTVDAMNVAALKIGEKSKSEREDIITEILDTLGLLDHRKTM TSG   637

Query   214  LSGGQQRKRLAIALELVNPPVMFFDEPTSGLDASCFQVVSLMKGLAQGGRSIICTIHQP   273
          LSGGQ+KRL+IALELV+NPP+MFFDEPTSGLD+SCFQ +SL+K LA+GGR+IICTIHQP
Sbjct   638  LSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQP   817

Query   274  SAKLFELFDQLYVLSQGQCVYRGKVCNLVPYLRDLGLNCPTYHN---PADFVMEVASGEY   330
          SA+LFE+FD LY L+ GQCVY+G      LVP+L  LG   P  H      P  +++ VA GEY
Sbjct   818  SARLFEMFDHLYTLADGQCVYQGSTKQLVPFLATLG--TPMXHRTTTPXSYIIXVAXGEY   991

Query   331  GD      332
          GD
Sbjct   992  GD      997
```

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result.

If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

[Q3] Gather information about this “novel” **protein**. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

```
>Tribolium castaneum protein (sequence taken from BLAST result)
NKMATDTVLLKNRSSDYKVQITPTQQKTLTHLPQRPKVDLAFSNLKYTVKQG-----
KA EK TILKSVSGRLRSGELTAIMGPSGAGKSTLLNLTGYRTDGVQGQILMNDSERDLSQFRKLSAYIMQ
DNQLHLNLTVD EAMNVA AKLKIG EKSKSEREDIITEILD TLGLLDHRKTM TSGLSGGQKKRLSIALELVS
NPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGR TI ICTI HQPSARLFEMFDHLYTLADGQCVYQGSTKQ
LVPFLATLG--TPMXHRTTXPXS YI IXVAXGEYGD
```

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as *S. cerevisiae*, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: Tribolium castaneum transporter

Species: Tribolium castaneum, red flour beetle (full lineage: Eukaryota; Opisthokonta; Metazoa; Eumetazoa; Bilateria; Protostomia; Ecdysozoa; Panarthropoda; Arthropoda; Mandibulata; Pancrustacea; Altocrustacea; Allotricarida; Hexapoda; Insecta; Dicondylia; Pterygota; Neoptera; Eumetabola; Endopterygota; Aparaglossata; Neuropteroidea; Coleoptera; Polyphaga; Cucujiformia; Tenebrionoidea; Tenebrionidae; Tenebrionidae incertae sedis; Tribolium)

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

- If there is a match with 100% amino acid identity to a protein in the database, from the same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.
- If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.
- If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.

- If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:

A BLASTP search, with the *Tribolium castaneum* protein sequence reported in Q2 as the query sequence, was run against the NR database (see first screenshot below).

The second screenshot shows the top hits from the search, with the max score belonging to an ABCG Member 4 protein from *Tribolium castaneum* (red flour beetle) at 93.56% identity, 99% coverage, and an E-value of 0.0 (accession number XP_001813184.1). Note that my original protein query from Question 1 was an isoform of *Homo sapiens* ABCG Member 1. The less-than-100% identity suggests a novel gene was found.

The third screenshot shows alignment details for the top hit discussed above, as well as the second highest scoring hit.

NIH National Library of Medicine
National Center for Biotechnology Information

BLAST® » blastp suite

Standard Protein BLAST

blastn blastp blastx tblastn tblastx

BLAST programs search protein databases using a protein query more...

Enter Query Sequence

Enter accession number(s), gis(s), or FASTA sequence(s) [Clear](#)

>Tribolium castaneum protein (sequence taken from BLAST result)
 NKKMATDTVLKNNSSDYKQITPTQOKTLHLPPRPKVDLAFNLKITYVQGG...
 KAEKTLKSVSGRLRSGLTAMPSGAGKSTLNLTYRTDGVGGQILMNDSE
 RDLGSPRLSAYFMQDQGLKALTYGSAHNAALKQKPSREDEITELDTL
 GLLDHVKMTSLGSGQKRLSIALELVNPPMFFDEPTSLDESSECFQCLL
 KTLARGRTICTGTHQPSAARLFEMFHYTLADGGVYGGSTKGLVPLATLG-
 TTMASHTTTPSRSTVAVGKSTVD

Query subrange

From To

Or, upload file No file chosen

Job Title Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database [Add organism](#)

Organism [Add organism](#)

Exclude ☐ Models (MMXP) ☐ Non-redundant RefSeq proteins (WPI) ☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm

Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

Note: Parameter values that differ from the default are highlighted in yellow and marked with + sign

+ Algorithm parameters

FOLLOW NCBI

< Edit Search

Save Search

Search Summary >

How to read this report?

BLAST Help Videos

Back to Traditional Results Page

Job Title

Tribolium castaneum protein (sequence taken...

RIID

YC495SJK014

Search expires on 04-21 19:42 pm

Download All >

Program

BLASTP

Citation >

Database

nr

See details >

Query ID

IciliQuery_1669409

Description

Tribolium castaneum protein (sequence taken from BLAS...

Molecule type

amino acid

Query Length

295

Other reports

Distance tree of results

Multiple alignment

MSA viewer >

Filter Results

Organism

only top 20 will appear

☐ exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity

E value

Query Coverage

to

to

to

Filter

Reset

Compare these results against the new Clustered nr database > BLAST

Descriptions	Graphic Summary	Alignments	Taxonomy
Sequences producing significant alignments			
Download Select columns Show 100 >			
☒ select all 100 sequences selected	GenPept	Graphics	Distance tree of results
		Multiple alignment	MSA Viewer
Description	Scientific Name	Max Score	Total Score
☒ ATP-binding cassette sub-family G member 4 [Tribolium castaneum]	Tribolium castaneum	596	556
☒ ATP-binding cassette sub-family G member 4 [Tribolium madens]	Tribolium madens	582	552
☒ hypothetical protein Zmor_024016 [Zophobas morio]	Zophobas morio	547	547
☒ unnamed protein product [Tenebrio molitor]	Tenebrio molitor	543	543
☒ ATP-binding cassette sub-family G member 4 [Zophobas morio]	Zophobas morio	533	533
☒ ATP-binding cassette sub-family G member 4 [Tenebrio molitor]	Tenebrio molitor	531	531
☒ hypothetical protein GEV33_008778 [Tenebrio molitor]	Tenebrio molitor	506	506
☒ hypothetical protein NQ215_007893 [Exocoetna adhaerens]	Exocoetna adhaerens	513	513
☒ unnamed protein product [Acanthoscelides obtectus]	Acanthoscelides obtectus	501	501
☒ ATP-binding cassette sub-family G member 4 isoform X1 [Anopheles gambiae]	Anopheles gambiae	506	506
☒ unnamed protein product [Callosobruchus maculatus]	Callosobruchus maculatus	499	499
☒ ABC subfamily G member 1 [Rhyzopertha dominica]	Rhyzopertha dominica	503	503
☒ unnamed protein product [Acanthoscelides obtectus]	Acanthoscelides obtectus	501	501
☒ hypothetical protein JTB14_036683 [Goniodroma autumnaeuncata]	Goniodroma autumnaeuncata	496	496
☒ ATP-binding cassette sub-family G member 1 [Leontotarsa decemlineata]	Leontotarsa decemlineata	494	494
☒ unnamed protein product [Phaedon cochleariae]	Phaedon cochleariae	494	494
☒ unnamed protein product [Callosobruchus maculatus]	Callosobruchus maculatus	489	489
☒ ATP-binding cassette sub-family G member 4 isoform X2 [Anopheles gambiae]	Anopheles gambiae	493	493

Alignment view Pairwise > Restore defaults > Download >

100 sequences selected >

Download > GenPept Graphics

Next > Previous < Descriptions

ATP-binding cassette sub-family G member 4 [Tribolium castaneum]

Sequence ID: XP_001813184.1 Length: 671 Number of Matches: 1

See 1 more title(s) > See all identical Proteins (PG)

Range 1: 1 to 292 GenPept Graphics

Next Match > Previous Match

Score	Expect	Method	Identities	Positives	Gaps
552 bits(1432)	0.0	Compositional matrix adjust.	276/295(94%)	278/295(94%)	5/295(1%)
Query 3	MATDTVLKRRSDYKVIPTPTQOKTLTHLPORPKVOLFANSLKYTVKGGAEKILKSV	62			
Sbjct 1	MGRNFLNNRSDYKVIPTPTQOKTLTHLPORPKVOLFANSLKYTVKGGAEKILKSV	60			
Query 63	SGRLRSGLTAINGPSGAGKSTLNLITGYRTDGVGGILLNDSERDLSQFRKL SAYDMQ	122			
Sbjct 61	SGRLRSGLTAINGPSGAGKSTLNLITGYRTDGVGGILLNDSERDLSQFRKL SAYDMQ	120			
Query 123	DNQLHNLTVDEAMVAALKIGEKSKSEREDIIITELDTLGLDHRKMTSGLSGGQKK	182			
Sbjct 121	DNQLHNLTVDEAMVAALKIGEKSKSEREDIIITELDTLGLDHRKMTSGLSGGQKK	180			
Query 183	RLSTALELVNPPIMFDFEPTSGLDSSSCFCISLTKTLARGRTICTTHOPSARLFEM	242			
Sbjct 181	RLSTALELVNPPIMFDFEPTSGLDSSSCFCISLTKTLARGRTICTTHOPSARLFEM	240			
Query 243	FDHLYTLADGGCVYGSTKQLVPFLATLG---TPMHRRTXPPSYIIXVAGEYGD	295			
Sbjct 241	FDHLYTLADGGCVYGSTKQLVPFLATLGLQCPSYHN---PASVIEVAGEYGD	292			

Download > GenPept Graphics

Next > Previous < Descriptions

ATP-binding cassette sub-family G member 4 [Tribolium madens]

Sequence ID: XP_044255746.1 Length: 671 Number of Matches: 1

Range 1: 1 to 292 GenPept Graphics

Next Match > Previous Match

Score	Expect	Method	Identities	Positives	Gaps
552 bits(1432)	0.0	Compositional matrix adjust.	273/295(93%)	278/295(94%)	5/295(1%)
Query 3	MATDTVLKRRSDYKVIPTPTQOKTLTHLPORPKVOLFANSLKYTVKGGAEKILKSV	62			
Sbjct 1	MGRNFLNNRSDYKVIPTPTQOKTLTHLPORPKVOLFANSLKYTVKGGAEKILKSV	60			
Query 63	SGRLRSGLTAINGPSGAGKSTLNLITGYRTDGVGGILLNDSERDLSQFRKL SAYDMQ	122			
Sbjct 61	SGRLRSGLTAINGPSGAGKSTLNLITGYRTDGVGGILLNDSERDLSQFRKL SAYDMQ	120			
Query 123	DNQLHNLTVDEAMVAALKIGEKSKSEREDIIITELDTLGLDHRKMTSGLSGGQKK	182			
Sbjct 121	DNQLHNLTVDEAMVAALKIGOKASEREDIIITELDTLGLDHRKMTSGLSGGQKK	180			
Query 183	RLSTALELVNPPIMFDFEPTSGLDSSSCFCISLTKTLARGRTICTTHOPSARLFEM	242			
Sbjct 181	RLSTALELVNPPIMFDFEPTSGLDSSSCFCISLTKTLARGRTICTTHOPSARLFEM	240			
Query 243	FDHLYTLADGGCVYGSTKQLVPFLATLG---TPMHRRTXPPSYIIXVAGEYGD	295			
Sbjct 241	FDHLYTLADGGCVYGSTKQLVPFLATLGLQCPSYHN---PASVIEVAGEYGD	292			

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width.

Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled sequences, for alignment:

*Note: FASTA header for novel protein was renamed from ">Tribolium castaneum protein (sequence taken from BLAST result)" to ">Red_flour_beetle protein (sequence taken from BLAST result)", to use the species' common name.

```
>Human gi|9619|refseq|NP_997512.1|ATP-binding cassette sub-family G member 1 isoform 7 [Homo sapiens]
NMEATETDLLNGHLKKVDNNLTEAQRFSSLPRAAVNIEFRDLSYSVPEGPWRRKKGYKTLLKGISGKFNSGELVAI
MGPSGAGKSTLMNILAGYRETGMKGAVLINGLPRDLRCFRKVSCYIMQDDMLLPHLTVQEAMMVSAHLKLQEKDEGR
REMVKEILTALGLLSCANTRTGSLSGGQQRKRLAIALELVNPPVMMFFDEPTSGLDASCFQVVSMLMKGLAQGGRSII
CTIHQPSAKLFELFDQLYVLSQGGQCVYRGKVCNLVPYLRDLGLNCPTYHNPADFVMEVASGEYGD
```

```
>Red_flour_beetle protein (sequence taken from BLAST result)
NKMATDTVLLKNRSSDYKVQITPTQQKTLTHLPQRPKVDLAFSNLKYTVKQGKAECTILKSVSGRLRSGELTAIMGPSG
SGAGKSTLLNILTGYRTDGVQGGQILMNDSEDLQFRKLSAYIMQDNQLHLNLTVDAMNVAALKIKIGKSKSERED
IITEILDTLGLLDHRKTMTSGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICT
IHQPSARLFEMFDHLYTLADGQCVYQGSGTKQLVPFLATLGTMPXHRTTTPXSYIIXVAXGEYGD
```

```
>Black_flour_beetle gi|123005823|refseq|XP_044255746.1|ATP-binding cassette sub-family G member 4 [Tribolium madens]
MGRNFLNNNRSSDYKVQITPTQQKTLTHLPQRPKVDLAFSNLKYTVKQGKAECTILKSVSGRLRSGELTAIMGPSG
AGKSTLLNILTGYRTDGVQGGQILMNDSEDLQFRKLSAYIMQDNQLHLNLTVDAMNVAALKIKIGDKAKSEREDII
TEILDTLGLLDHRKTMTSGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIH
QPSARLFEMFDHLYTLADGQCVYQGSGTKELVPFLATLGLQCPHYHNPASYIIEVACGEYGD
```

```
>Superworm gi|135128968|refseq|XP_063912122.1|ATP-binding cassette sub-family G member 4 [Zophobas morio]
MGRNFLNNNRSSDYKVQITPAQPKTLTHLPQRPKVDLAFSNLKYTVKQGKVEKTLKSVSGRLRSGELTAIMGPSG
AGKSTLLNILTGYRTDGVQGGQILMNDNERDLQFRKLSAYIMQDNQLHLNLTVDAMNVAALKIKIGDKPKSERAII
HEILDTLGLLDHRKTMTSGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIH
QPSARLFEMFDHLYTLADGQCVYQGSGTKQLVPFLASLSLQCPAYHNPASYIIEVACGEYGD
```

```
>Yellow_mealworm gi|138133997|refseq|XP_068908145.1|gb|KAJ3631279.1|ATP-binding cassette sub-family G member 4 [Tenebrio molitor]
```


MGRNFLNNNRSSDYKVQIQPLQOKTLTHLPQRPKVDLAFSNLNYTVKQGKVEKTILKSVSGRLRSGELTAIMGPSG
AGKSTLLNILTGYRTEGVKGQILMNDSELDLSQFRKLSAYIMQDNQLHLNLTVDAMNVAALKKIGDKPKSERDAII
TEILDTLGLSDHRKTMSTGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIH
QPSARLFEMFDHLYTLADGQCVCYQGSTNQVLVPFLATLGLQCPSYHNPAASYIIIEVACGEYGD

>Asian_longhorned_beetle gi|108906469|refseq|XP_018565251.1|ATP-binding
cassette sub-family G member 4 isoform X1 [Anoplophora glabripennis]
LATDNLLLKNRSPDVKVQITPAQPKKLTHLPERPKVDLAFSNLTYYTVRQGGKDKIILKNVSGRLRSGELCAIMGPSG
AGKSTLLNILTGYKTEGTTGQILMNNNERDLSQFRKLSAYIMQDNQLHANLTVDEAMTVAALKKIGDKKLHDREAII
TEILDTLGLLDHRKTMSTGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLAKGGRTIICTIH
QPSARLFEMFDQLYTLADGQCVCYQGSTNQVLVPFLGTLGLQCPSYHNPAASYIIIEVACGEYGD

>Lesser_grain_borer gb|KAI7815555.1|ABC subfamily G member 1 [Rhyzopertha
dominica]
ELATDTLLLKNRGSDAVKVQITPCQTKTLTHLPRRPKVDLAFENLTYYTVKQGKNDKTILKSVSGRLRSGELTAIMG
SGAGKSTLLNILTGYKTEGLSGTILMNNERDLSQFRKLSAYIMQDNQLHANLTVDEAMNVAALKKIGDKKSHAREA
IITEILDTLGLLDHRRTMTSSLSGGQKKRLSIALELVSNPPVMMFFDEPTSGLDSSSCFQCISLLKTLAGGGRTIICT
IHQPSARLFEMFDQLYTLSEGQCVCYQGSTKQLVPFLGTLGLQCPSYHNPAASFIIIEVACGEYGD

>Colorado_potato_beetle gi|111514950|refseq|XP_018565251.1|ATP-binding
cassette sub-family G member 1 [Leptinotarsa decemlineata]
MATDNLLLKTTSTDVKVQIVSTPKKNLTHLPRPKVDLAFSNLKYVVKQGKTEKVILRNVSGHIRSGELCAIMGPSG
AGKSTLLNILTGYKTEGMSGQILMNDADRDLDGQFRKLSAYIMQDNQLHANLTVDEAMSVAALKKIGDKPVHEREAVI
TEILDTLGLIDHRKTMSTGLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLAKGGRTIICTIH
QPSARLFEMFDQLYTLADGQCVCYQGSTNQVLVPFLGSLGLQCPSYHNPAASYIIIEVACGEYGD

Alignment:

Obtained using MUSCLE (version 3.8) at EBI:

CLUSTAL multiple sequence alignment by MUSCLE (3.8)

Human	NMEATETDLLNGHL--KKVDNNLTEAQRFFSLPRRAAVNIEFRDLSYSVPEGPWWRRKKG
Red_flour_beetle	NKMATDTVLLKNRSSD-YKVQITPTQOKTLTHLPQRPKVDLAFSNLKYTVKQG-----KA
Superworm	--MGRNFLNNNRSSD-YKVQITPAQPKKLTHLPQRPKVDLAFSNLKYTVKQG-----KV
Yellow_mealworm	--MGRNFLNNNRSSD-YKVQIQPLQOKTLTHLPQRPKVDLAFSNLNYTVKQG-----KV
Black_flour_beetle	--MGRNFLNNNRSSD-YKVQITPTQOKTLTHLPQRPKVDLAFSNLKYTVKQG-----KA
Colorado_potato_beetle	--MATDNLLKTTSTD-VKVQIVSTPKKNLTHLPRPKVDLAFSNLKYVVKQG-----KT
Asian_longhorned_beetle	--LATDNLLLKNRSPD-VKVQITPAQPKKLTHLPERPKVDLAFSNLTYYTVRQG-----KK
Lesser_grain_borer	-ELATDTLLLKNRGSDAVKVQITPCQTKTLTHLPRRPKVDLAFENLTYYTVKQG-----KN
	. : * : ** : : : * * . * : * : * * * : *

Human	YKTLKGISGKFNSGELVAIMGPSGAGKSTLMNILAGYRETGMKGAVLINGLPRDLRCFR
Red_flour_beetle	EKTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYRTDGVQGQILMNDSELDLSQFR
Superworm	EKTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYRTDGVQGQILMNDSELDLSQFR
Yellow_mealworm	EKTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYRTEGVKGQILMNDSELDLSQFR
Black_flour_beetle	EKTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYRTDGVQGQILMNDSELDLSQFR
Colorado_potato_beetle	EKVILRNVSGHIRSGELCAIMGPSGAGKSTLLNILTGYKTEGMSGQILMNDADRDLDGQFR
Asian_longhorned_beetle	DKIILKNVSGRLRSGELCAIMGPSGAGKSTLLNILTGYKTEGTTGQILMNNNERDLSQFR
Lesser_grain_borer	DKTILKSVSGRLRSGELTAIMGPSGAGKSTLLNILTGYKTEGLSGTILMNNERDLSQFR
	* : * . : * * . : *

Human	KVSCYIMQDDMLLPHLTQVEAMMVAHLKLQEKDEGRE-MVKEILTALGLLSCANTRTG
Red_flour_beetle	KLSAYIMQDNQLHLNLTVDAMNVAALKKIGEKSKSEREDIITEILDTLGLLDHRKTMST
Superworm	KLSAYIMQDNQLHLNLTVDAMNVAALKKIGDKPKSERAIIHEILDTLGLLDHRKTMST
Yellow_mealworm	KLSAYIMQDNQLHLNLTVDAMNVAALKKIGDKPKSERDAIITEILDTLGLSDHRKTMST
Black_flour_beetle	KLSAYIMQDNQLHLNLTVDAMNVAALKKIGDKAKSEREDIITEILDTLGLLDHRKTMST
Colorado_potato_beetle	KLSAYIMQDNQLHANLTVDEAMNVAALKKIGDKPVHEREAVITEILDTLGLIDHRKTMST
Asian_longhorned_beetle	KLSAYIMQDNQLHANLTVDEAMTVAALKKIGDKKLHDREAIIITEILDTLGLLDHRKTMST
Lesser_grain_borer	KLSAYIMQDNQLHANLTVDEAMNVAALKKIGDKKSHAREAIIITEILDTLGLLDHRKTMST

```

*:*.*****: * :***:*** *:***: :* * :. *** :*** . .* *.

Human          SLGGGQRKRLAIALELVNPPVMFFDEPTSGLDSSCFQVVSMLMKGLAQGGRSIICTIHQ
Red_flour_beetle GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Superworm      GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Yellow_mealworm GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Black_flour_beetle GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Colorado_potato_beetle GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Asian_longhorned_beetle GLSGGQKKRLSIALELVSNPPIMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
Lesser_grain_borer SLSGGQKKRLSIALELVSNPPVMFFDEPTSGLDSSSCFQCISLLKTLARGGRTIICTIHQ
.*****.***:*****.***:*****:**** :*: * ** ***:*****

Human          PSAKLFELFDQLYVLSQGCQVYRGKVCNLPYLRDLGLNCPTYH--NPADFVMEVASGE
Red_flour_beetle PSARLFEMFDHLYTLADGQCQVYQGSTKQLVPFLATLG--TPMXHRTTXPXSYIIXVAXGE
Superworm      PSARLFEMFDHLYTLADGQCQVYQGSTKQLVPFLASLSLQCPAYH--NPASYIIEVACGE
Yellow_mealworm PSARLFEMFDHLYTLADGQCQVYQGSTNQLVPFLATLGLQCPSYH--NPASYIIEVACGE
Black_flour_beetle PSARLFEMFDHLYTLADGQCQVYQGSTKELVPFLATLGLQCPSYH--NPASYIIEVACGE
Colorado_potato_beetle PSARLFEMFDQLYTLADGQCQVYQGSTNQLVPFLGSLGLQCPAYH--NPASYIIEVACGE
Asian_longhorned_beetle PSARLFEMFDQLYTLADGQCQVYQGSTNQLVPFLGTLGLQCPSYH--NPASYIIEVACGE
Lesser_grain_borer PSARLFEMFDQLYTLSEGQCQVYQGSTKQLVPFLGTLGLQCPSYH--NPASFIEVACGE
***.***:***:***.***:*****.*.. :***:* *. * * * * .::: ** **

Human          YGD
Red_flour_beetle YGD
Superworm      YGD
Yellow_mealworm YGD
Black_flour_beetle YGD
Colorado_potato_beetle YGD
Asian_longhorned_beetle YGD
Lesser_grain_borer YGD
***

```

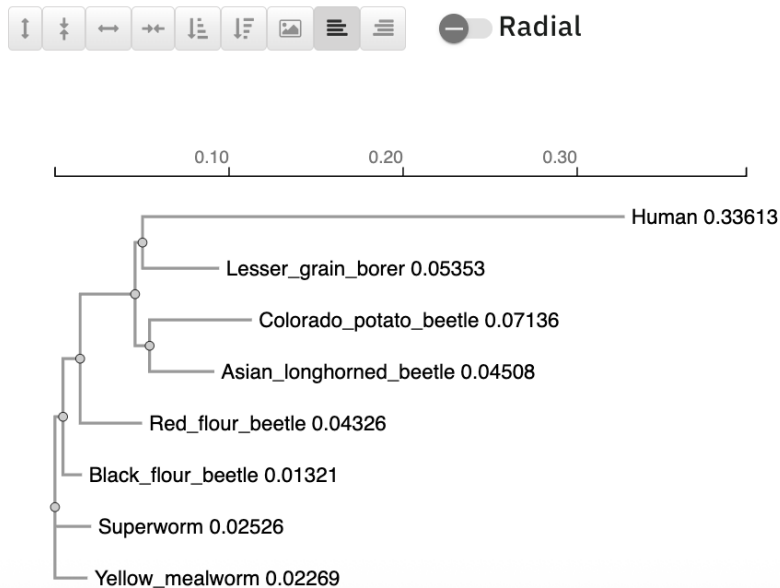
[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use “simple phylogeny” online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

The phylogenetic tree (see “Phylogram”, below) was generated using “**Simple Phylogeny**” from EBI, with MUSCLE-generated MSA file (CLUSTAL format) used as input.

Phylogenetic Tree

```
(
(
(
(
(
Human:0.33613,
Lesser_grain_borer:0.05353)
:0.00519,
(
Colorado_potato_beetle:0.07136,
Asian_longhorned_beetle:0.04508)
:0.01015)
:0.03831,
Red_flour_beetle:0.04326)
:0.01192,
Black_flour_beetle:0.01321)
:0.00562,
Superworm:0.02526,
Yellow_mealworm:0.02269);
```

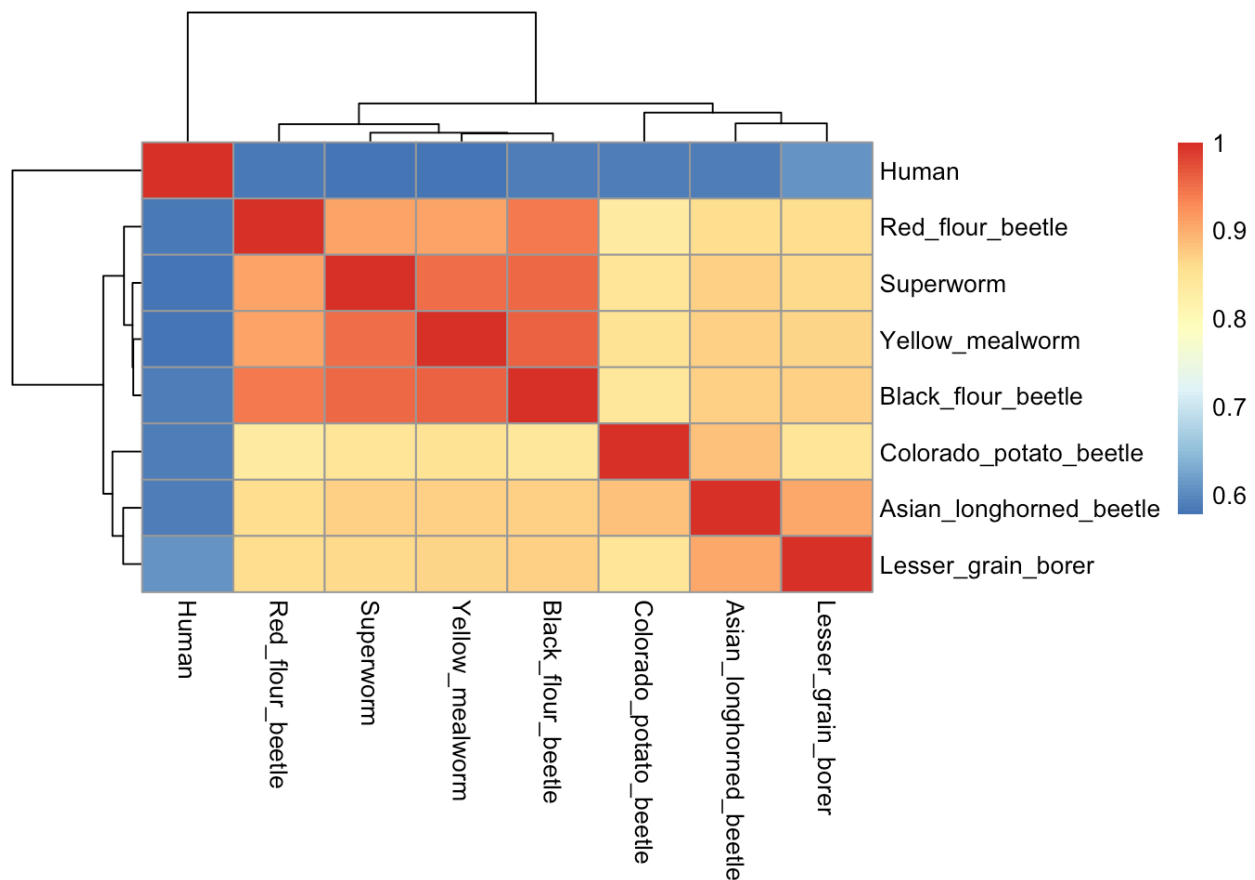
Phylogram



[Q7] Generate a sequence identity based **heatmap** of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and “Save as” FASTA format for example). Read this FASTA format alignment into R with the help of functions in the **Bio3D package**. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

Steps taken:

- FASTA format of sequence alignment downloaded directly from MUSCLE output (“The alignment in FASTA format converted by Seqret”)
- FASTA file read into R using “read.fasta()” function, in bio3d package
- Sequence identity matrix generated using “seqidentity()” function, in bio3d package
- Heatmap generated using output matrix and “pheatmap()” function, in pheatmap package



[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 *unique* hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function `consensus()`. The Bio3D functions `blast.pdb()`, `plot.blast()` and `pdb.annotate()` are likely to be of most relevance for completing this task. Note that the results of `blast.pdb()` contain the hits PDB identifier (or `pdb.id`) as well as Evalue and identity. The results of `pdb.annotate()` contain the other annotation terms noted above.

Note that if your consensus sequence has lots of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by calculating the row-wise maximum from your sequence identity matrix.

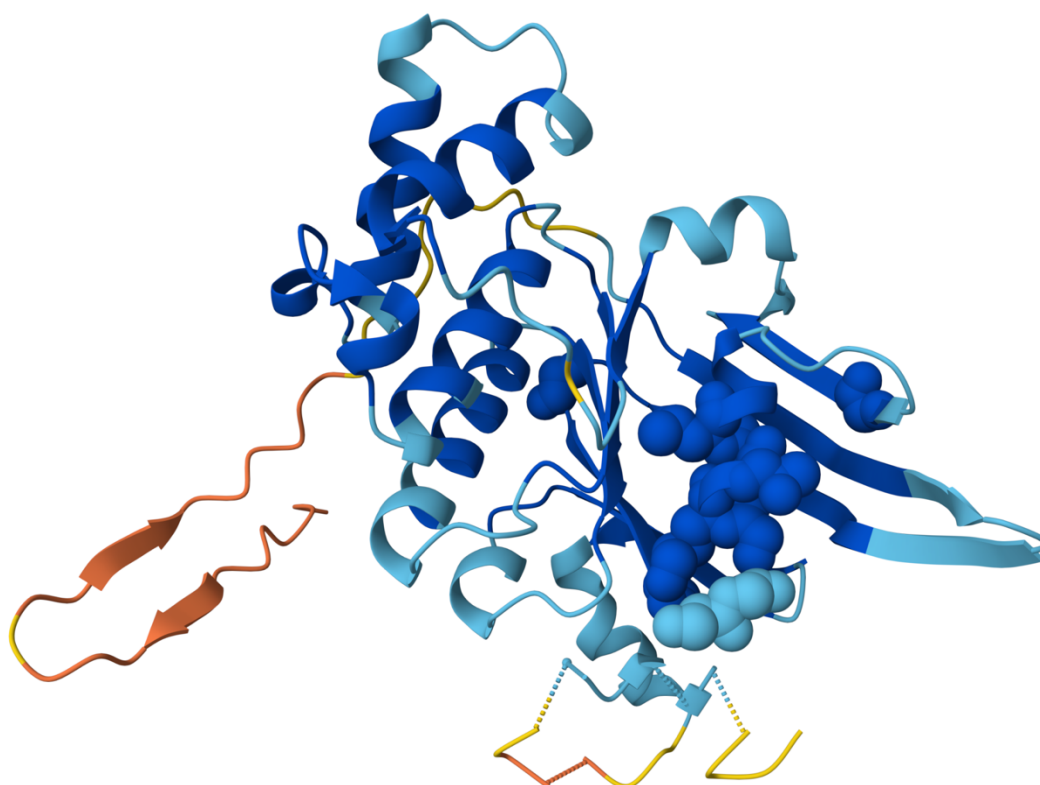
PDB ID	Technique	Resolution	Source	E-value	Identity
7R8C	EM (Electron microscopy)	3.70	Homo sapiens	8.50e-96	58.966
8I38	EM	2.90	Arabidopsis thaliana	3.32e-41	37.008
9IUK	EM	3.38	Candida albicans SC5314	1.75e-36	38.583

Note: The `plot.blast()` function returned only 4 hits, all of which were Homo sapien. The top Homo sapien hit is reported here (1st hit, 7R8C), as well as the next two highest `blast.pdb()` search hits from different species (5th hit, 8I38, and 15th hit, 9IUK).

[Q9] Using [AlphaFold notebook](#) generate a structural model using the default parameters for your novel protein sequence.

Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a “too many amino acids” (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for [PFAM](#) domain matches.

Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the **Mol* viewer** online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight conserved residues that are likely to be functional as **spacefill** and the protein as **cartoon** colored by local alpha fold *pLDDT* quality score. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that *pLDDT* score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).

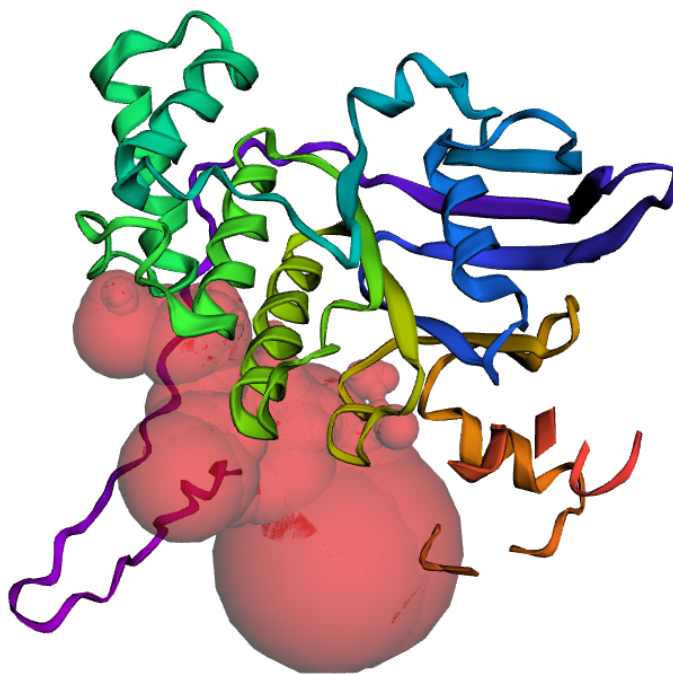


Relevant information:

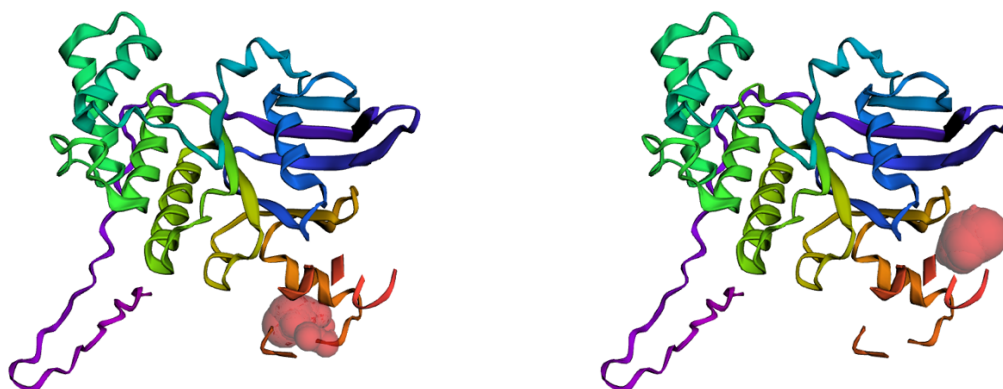
- Due to high diversity of my protein alignment, a **50% conservation cutoff score** was chosen, which identified the **ten Spacefill-represented residues** shown above as relatively most conserved (of 295 residues, in total).
- The polymer (including the Spacefill residues) is **colored by “pLDDT Confidence”** (under Validation, in Mol*), where blue represents very high quality and orange represents very low quality.

[Q10] (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (<https://cfold.bme.uic.edu/castpfold/>). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s **area** and **volume**.

The largest predicted pocket’s negative volume is shown in the image below, in red, with an **area** of 956.594 Å² and a **volume** of 2236.162 Å³.



The predicted pocket described and pictured above is substantially larger than the other predicted pockets. For comparison, the second largest pocket (below, left) has an area of 156.159 Å² and volume of 96.010 Å³, while the third largest pocket (below, right) has an area of 76.399 Å² and volume of 93.602 Å³.



(ii) Perform a “Target” search of ChEMBL (<https://www.ebi.ac.uk/chembl/>) with your novel sequence. Are there any **Target Associated Assays** and **ligand efficiency data** reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

Upon performing a “Target” search with the novel sequence, ChEMBL reported 88 targets, none of which are the query protein ABCG1. Thus, strictly speaking, it appears no target-associated assays or ligand efficiency data are available as of 06/07/2026.

The highest reported target (with an E-value of 1.3e-42 and only 37.4% identity) is [CHEMBL3509585](#): “Broad substrate specificity ATP-binding cassette transporter ABCG2” in *Rattus norvegicus*, with 28 ADME assays (measuring drug absorption, distribution, metabolism, and excretion) and no ligand efficiency data. Though ABCG2 is a member of the ABCG subfamily of ATP-binding cassette transporter proteins, it has been extensively reported that members of the ABCG family have very distinct structures and substrate specificities, owing to their different transport functions/mechanisms (Xu et al., 2022). Only ABCG1 and ABCG4 are considered structurally similar, as both are involved in lipid/sterol transport (whereas ABCG2 is a multidrug/xenobiotic transporter) – ABCG4 is also not among the search results.

A direct search in ChEMBL finds that ABCG1/4 are not in the database’s listed targets, suggesting there have been few assays exploring their inhibition that could be extended to the novel protein.

Xu, D., Li, Y., Yang, F., Sun, C.-R., Pan, J., Wang, L., Chen, Z.-P., Fang, S.-C., Yao, X., Hou, W.-T., Zhou, C.-Z., & Chen, Y. (2022). Structure and transport mechanism of the human cholesterol transporter ABCG1. *Yearbook of Paediatric Endocrinology*, 38(4). <https://doi.org/10.1530/ey.19.14.3>

(iii) Briefly discuss (100 words max) the **druggability** of your novel protein based on:

- Presence of well-defined pockets (output of tools like CASTpFold),

- Existence of known inhibitors for related proteins (your search of ChEMBL),
- Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

The novel protein currently has limited druggability. While 160-800 Å³ is optimal for small-molecule binding sites², CASTpFold's largest-identified pocket is poorly defined and overly large (2236 Å³), while the next largest are under 100 Å³. Furthermore, the protein is a ABCG1 homolog, which is not a ChEMBL-reported target and lacks known inhibitors; the closest ChEMBL-reported relative, ABCG2, has structurally distinct binding sites³. Finally, the MSA's most conserved residues are not located in CASTpFold-identified pockets and have only 50-70% conservation across homologs. Given its role in cholesterol efflux, targeting the protein could protect against cardiovascular disease risk factors such as atherosclerosis¹.

1. Hardy, L., Frisdal, E., & Le Goff, W. (2017). Critical Role of the Human ATP-Binding Cassette G1 Transporter in Cardiometabolic Diseases. *International Journal of Molecular Sciences*, 18(9), 1892. <https://doi.org/10.3390/ijms18091892>
2. Loving, K. A., Lin, A., & Cheng, A. C. (2014). Structure-Based Druggability Assessment of the Mammalian Structural Proteome with Inclusion of Light Protein Flexibility. *PLoS Computational Biology*, 10(7), e1003741. <https://doi.org/10.1371/journal.pcbi.1003741>
3. Xu, D., Li, Y., Yang, F., Sun, C.-R., Pan, J., Wang, L., Chen, Z.-P., Fang, S.-C., Yao, X., Hou, W.-T., Zhou, C.-Z., & Chen, Y. (2022). Structure and transport mechanism of the human cholesterol transporter ABCG1. *Yearbook of Paediatric Endocrinology*, 38(4). <https://doi.org/10.1530/ey.19.14.3>